



explain Detailed Design Review in detail and with substeps

Detailed Design Review (DDR) is the checkpoint where you verify that the design is fully engineered, documented, and ready for simulation, prototyping, and manufacturing—with risks understood and controlled. Below is a detailed breakdown into practical substeps.^[1] ^[2]

1. Confirm review objectives and readiness

- Define what this DDR must confirm: that the detailed design implements all requirements, is producible, safe, reliable, and ready for release to simulation/prototype or production data.^[3] ^[1]
- Check entry criteria: all key components modeled, drawings available, critical analyses done, major issues from PDR closed or with clear action plans.^[4] ^[5]

2. Review complete design definition (CAD and drawings)

- Inspect 3D CAD models of all parts and assemblies: geometry, constraints, clearances, interferences, motion, and assembly sequence feasibility.^[2] ^[6]
- Review 2D drawings (if used): views, dimensions, tolerances, surface finishes, GD&T, material specs, weld symbols, and notes are complete and consistent with company standards.^[6] ^[1]

3. Check requirements compliance

- Trace each requirement (functional, performance, regulatory, safety, reliability, environmental) to specific design features, calculations, or analyses that show how it is met.^[1] ^[3]
- Identify any requirements not fully satisfied, partially satisfied, or to be verified only in later testing, and document associated risks or waivers.^[5] ^[3]

4. Assess analyses and simulations

- Review detailed engineering analyses: strength, fatigue, thermal, vibration, fluid flow, kinematics, tolerance stack-ups, and worst-case conditions.^[7] ^[2]
- Confirm that safety factors, margins, and performance reserves meet internal guidelines and that analysis assumptions match the real design and use cases.^[7] ^[5]

5. Evaluate manufacturability and assembly (DFMA)

- Check that features, tolerances, and processes are suitable for the chosen manufacturing methods (machining, casting, injection molding, sheet metal, etc.).^{[2] [6]}
- Evaluate assembly: part count, ease of assembly, use of standard components, accessibility of fasteners, error-proofing (Poka-Yoke), and need for special tools or jigs.^{[6] [1]}

6. Review materials, components, and standards

- Verify material selections, treatments, and coatings with respect to strength, corrosion, wear, weight, cost, and availability.^{[7] [2]}
- Ensure use of preferred parts and standard hardware where possible, and check that any special or long-lead components have been identified and approved.^{[2] [6]}

7. Check interfaces and integration

- Confirm that mechanical, electrical, fluid, and software interfaces match corresponding subsystems and external systems (envelopes, mounting, connectors, port sizes, communication protocols).^{[1] [6]}
- Validate interface definition via interface control documents (ICDs), reference models, and integration diagrams, ensuring there are no overlaps, clashes, or inconsistent specs.^{[6] [2]}

8. Validate safety, reliability, and compliance aspects

- Review safety features (guards, interlocks, fail-safe behavior, emergency stops, warning labels) and confirm alignment with relevant norms/standards.^{[7] [6]}
- Assess reliability and maintainability design: access for service, component derating, redundancy, diagnostic features, and results from FMEA or similar analyses.^{[2] [7]}

9. Review documentation and configuration control

- Ensure all design documents are up to date and consistent: BOMs, drawings, CAD, specifications, interface documents, calculations, and test/inspection requirements.^{[3] [1]}
- Confirm that configuration management is in place: unique IDs, revision control, change history, and clear rules for future design changes after DDR.^{[4] [5]}

10. Examine project impacts: cost, schedule, and risk

- Review cost estimates based on the detailed design: material, manufacturing, assembly, tooling, testing, and lifecycle costs.^{[1] [2]}
- Update risk register with any design-driven risks (e.g., tight tolerances, new processes, supplier dependencies) and confirm contingency or mitigation plans.^{[5] [4]}

11. Conduct the review session

- Invite cross-functional participants (design, analysis, manufacturing, quality, supply chain, service, project management) and share the review package well in advance.^{[3] [2]}
- During the meeting, walk through the system and key subsystems, collect findings, and log actions with owners and due dates using checklists or structured templates.^{[8] [4]}

12. Decide outcome and approve for next phase

- Decide whether the design is: approved, approved with minor actions, conditionally approved pending major actions, or rejected requiring re-design and re-review.^{[5] [3]}
- If approved, formally release the design to the next phase (e.g., Simulation Review, prototype build, or manufacturing) and freeze the relevant baseline.^{[6] [2]}

13. Record lessons learned and improvements

- Document key issues, root causes, and good practices identified during the DDR so they can improve future projects and checklists.^{[4] [5]}
- Update your design review templates and standards based on what worked well or poorly in this review cycle.^{[8] [4]}

If you'd like, I can now compress these DDR substeps into a yes/no checklist specifically tuned for mechanical designs (clearances, fits, GD&T, DFMA, safety) that you can drop straight into your app.

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9. <https://www.scribd.com/document/466004378/DR3-Detailed-Design-Review-Guidelines>

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